BUILDING DESIGN TOOL ARTEFACTS AS PERSONAL THEORY

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ABSTRACT

This paper describes a practice-led research project that has proceeded through building three different design tool artefacts. The tools were built on the assumption that design tools are seeds for design ideas and guide their formation. Material design tools and computer based tools both work as bases about how and what to design. Instead of seeing them as just assistive devices, they may be used for setting the stage for design in the first place. The three tools, designed artefacts in themselves, are described as stemming from initial premises into concepts that have generative potential. The process of building these tools are discussed in terms of practitioner knowledge. As a method, engagement into tool building is a self-imposed contract through which also theoretical insight and learning becomes generated.

Keywords: practice-led research, design tools, computer tools

INTRODUCTION

Practice-led and practice-based research approaches have been advocated by artists and designers who have wanted to include design work as part of their research project. By reporting on the practical design work he or she has done, the designing researcher is contributing to the wider repertoire of design knowledge. Looking at the process of making tools as a part of a personal theory building process is a less explored angle.

The research question has been how designers could have tools that are more attuned to their personal beliefs, and what this would benefit them. Personal theory here means personal philosophies and artistic credos. The designer, working creatively, is not likely to use a combination of well-understood academic theory and a practical skill. The designer's personal knowledge is a combination of literature read, influences and experiences of done things. In this light, tool building becomes also a way of personal theory building.

As a graduate from a furniture and interior design program, I became interested in the way the available tools and materials can be suggestive in the idea-generating phases of design. Past experience with woodcraft has also helped appreciate the ways not only materials but available machine workshop tools suggest outcomes. Acquired skills with drawing and programming also stimulated me to turn toward computer programming and drawing for examining the role of tools in a generative, creative process.

The research was motivated by interest in the possibilities of computer tools in design, and thus becomes positioned along recent research into computer tools emerging from designing researchers. As it is difficult for many designers to literally build their own software tools, it is worthwhile to ask how a designer might utilize computers in a personal design approach.

The method of approach has been to research the topic in a practice-led manner, allowing practical design projects to influence and inform the research process and reading of literature. In this project I have built three definite design artefacts. The writing of the research partly a report of the design process, but also includes discussing the emerging themes in dialogue with existing research literature and precedents.

BACKGROUND

In this research project, tools are brought to foreground as both designed objects and as the originators of further design tool ideas. The tools, as designed objects, can be interpreted in a material and conceptual role. I consider practice-based research and designers' own research into design tools and materials to be a precedent to this project.

Designers produce new things and also associated new knowledge. Simply from a technical standpoint, making objects can prove assumptions, such as successful use of new material for a chair. This kind of claim of technical novelty does not generally interest researchers in art and design context. Artistically-oriented designers initiate projects which demonstrate experimental work that addresses theoretical, conceptual or philosophical questions worthy of academic research.

For example, Nithikul Nimkulrat (2009), a textile artist, who worked with paper material in her thesis project, had no primary interest in the technical properties, but the expressive potential of the paper material. Maarit Mäkelä's (2003) work with clay does not centrally attempt to prove any new method or technique of working with clay, but instead offers a way of discussing femininity and representations of gender.

I interpret this type of research project as partly a learning process, following Donald Schön's definition of a reflective practice. (1991) Research is not just learning a new fact or knowledge about the world, but learning new how-to, skill or extensions to old skills. Distributing this knowledge is reporting the process in a way that can demonstrate to others a means to improve their skills and knowledge in similar manner.

Similarly, this means the outcomes and findings of this kind of research process are not definite facts. The process can in aspects be replicated by someone else. The write-up of the project ought to show clearly in what way the research and design was built up, so that not only the design work sets an example but the overall process works as an example of a method of approach.

FOCUS ON GENERATIVE TOOLS

This project has focused on generative aspects of tools, making generation a central concept for understanding the significance of design tools, both material and conceptual. It is also used to bridge the relation between tools and personal design theories.

Generative aspects of art were already discussed by the artist and teacher Paul Klee in the 1920s. (Klee, 1961) Klee's lines on paper, starting from a single point, begin to play each off on the surface, producing images through the organization the artist reasons with. In this way, the draftsman does not transfer a fully formed idea into something visible, but also allows the drawing to lead. This attitude to drawing delineates the generative role of the drawing from other purposes. For Klee, genesis of a work is the account of how a work of art came into being. The genesis of the drawing may be explained and shared, as it is not necessarily visible in the outcome.

Theorists of design have also examined the presence of generative activities in creative design processes. Herbert Simon suggested that styles of design would originate from different choices of approach, each a preference rather than necessity. For example, the choice of designing a house from inside out instead of outside in, would presumably result in a different style. Whole schools of design could emerge from differences like this. (Simon, 1975)(See also Simon, 1996, 128-130) Donald Schön revisits this same point in his book *The reflective practitioner,* noting that these moves are less obvious in the hands of more masterful designers. (Schön, 1991, 103-104) Schön also discussed the way certain words may work as generative metaphors, not only as a way of seeing a thing as something else but influencing the framing and the solution of the problem itself. (Ibid, 184-187).

Christopher Alexander discussed the designer's education as significant in building the worldview and schemas from which to derive design ideas. At the same time he is critical of invented concepts and general principles fuelled by architectural "theory". To Alexander, adoption of geometries and organizing

devices appeared as lazy moves to avoid modeling real world complex problems. (Alexander, 1964, 62-63.)

Jane Darke (1984) introduced the concept of a primary generator and generative strategy, the terminology which is followed here. (See also Lawson, 2006, 188-195.) These concepts are useful in putting together both the generative process and the choice element of making the decision in the first place. A visual identity for the design project may become as a flash of insight and remain the guiding light throughout the process. The generative strategy is something that at the same time makes the design task at hand more manageable and sets it up creatively and gives it direction.

The view here is that tools can either impose these approaches, suggest them, or they can be flexible and rich as to allow the building of generative approaches and rules within them. Klee's understanding of drawing is an example of such a broad medium. In contrast, a perspective method on paper can be a rigid application of a set of rules, acted out through the use of mechanical rulers and auxiliary lines. But perspective can also remain a flexible idea for the person who does the sketching, without applying a rigid method, just as Klee did with his instructive drawings.

The generative strategies are major components of personal, subjective theories of design. Continuous use of the same material or medium can provide a basis for the beginnings of each new art piece. Attachment to a particular material or tools is tied to personal preference or a maturing personal philosophy of the artist. In design, it may be more common to use methods of approach, tools and representations, rather than one specific material medium. Yet this is by no means an essential difference between designers and artists, as both avenues are open for the different practitioners. An artist can exercise distance from the actual materials, whereas an architect might arrive at form through very tangible and intimate relation with a chosen intermediate material. Again, a particular approach arises from preference and a cultivation of a personal belief.

An artistic credo or a belief system is thus an important base for suggesting new outcomes arising from underlying design philosophy. Systems of harmonious proportions or the classical orders of architecture are examples of past ideas which were very widespread and also transmitted within the discipline. Such ideas are not valid as descriptive or predictive general theory of design, but are instead material for personal theory-building and learning.

In the present study, generation is seen as something both chosen and maintained by a person. This is different from purely algorithmic generation, which often is used to detach the designer's direct influence from the process. The first artefact touches on this topic but is ultimately seen as an attempt at personal theory construction and not algorithmic form generation.

TOOLS AND PRACTICE-LED RESEARCH

I consider Caroline Hummels' work into gestural design tools (2000) and Birger Sevaldson's work into digital design techniques (2005) to have influenced the outlining of this project during its course. Sevaldson states that his thesis is practice-based, meaning that it is an extension of his diverse work as a practicing designer over the years. The practice is closer to the field of architecture than for example industrial design. Hummels is not explicitly stating her research to be a practice-based or practice-led, but she clearly makes use of design activity.

Hummels supports a participatory approach, where the more commonly shared motions of gestures, dance, and other experiential, tangible aspects are inspected as potential future design tools both in generative and descriptive capacity. Sevaldson instead considers algorithmic generation and use of various datasets primarily from the designer's viewpoint. Among other things, visual form generation becomes rethought in using computers, which can help break established schemata and create emergent material, unexpected surprises. The active eye of the designer is able to distinguish the interesting from the on-screen material.

Importantly, both projects are clearly insider views into design tools, emerging from design fields. Yet the two works differ both ideologically and in the way they approach the topic, suggesting the works have a basis in a developed personal credo.

METHOD AND APPROACH

I chose, as part of my method, to build design tools within a research process. The design work has started from a belief that tools of design, especially computer tools nowadays, ought to be studied and developed by designers themselves. In context of practice-led research, the building of tools as a central topic is a less explored angle.

The present research sees design tools as a part of practitioner knowledge. Practitioners have always built tools for general utility or particular situations as needs arise. In craft tradition tool knowledge could be disseminated informally or through the master-apprenticeship relation. A modern perspective method manual is an example of a more recent tradition of disseminating useful tool knowledge, emerging from the design practitioners, for the practitioners.

For defining the research method and mode of outcomes I've found useful Donald Schön's suggestions of how a practitioner could engage in research in more systematic ways. This "reflective research" can be undertaken to enhance the practitioner's reflection-in-action. (Schön, 1991, 309) Repertoire building is mentioned as one approach. Lawyers have their legal cases and architects are familiar with precedents. (Ibid, 309-317) To go past the superficial, the precedent description benefits from an account of how problems emerged and how they were solved. The thinking that informed choices ought to be made visible when possible.

Design artefacts itself are an important aspect of the repertoire building. Biggs (2002) has argued that artefacts alone do not work as a research contribution. This is in line with Schön's suggestion. Text and artefacts together becomes a fully formed research outcome. For Biggs, including an account of a context completes the artefacts as distributable knowledge. Otherwise the knowledge remains too subjective, just as works of art in an exhibition are open to multiple interpretations. For design and art alone, the multitude of interpretations can be even desirable. Research outcomes ought to be less ambiguous. Yet it has to be stressed that in research, accompanying artworks with text is not to give the audience an interpretation of the work, but to explain the activities that were relevant for the genesis of the work.

As regards the method of approach, I see the thesis works by Mäkelä (2003) and Nimkulrat (2009) valuable. They have described more precisely the way their practical work intertwines with the theoretical and literature parts. The interaction between practice parts and their reflective analysis is something that when presented together allows their examination and to an extent, replication. Each successive art exhibition became a milestone, which suggested more dialogue with literature and looking back at the work done.

The practitioner has accumulated skills and knowledge which at a given time forms a personal repertoire. This involves both experiences and things seen. (Schön, 1991, 138) This kind of knowledge goes beyond the individual cases. Looking back at the knowledge and experiences retrospectively, the knowledge can be reported to others in a way that is useful to a community of practitioners and researchers. The intelligent reporting of practical works builds a more shared repertoire of knowledge, accessible to others.

This research has in part allowed the practical work to lead the research project and the reading of theory. The text is an account of looking back at the making of the tools and the motives and impetus that preceded them.

THE THREE ARTEFACTS

This part summarizes the three artefacts that were specifically used as case studies in the research project. Each probes into the general topic of how space and form is understood from a design viewpoint. The tools are presented in the order they were made, as the earlier informed the later.

The first artefact is a visualization of the experience of motion. The second artefact explores the notion of a tool as a hand-held device and building a conceptual strategy out of it. The third addresses the topic of directly making spatial form, by creating a modelling program and relating it to the sketching process that paralleled it.

The two first artefacts are significant steps in the exploratory stage of the research, whereas the work around the third artefact forms the more central proposal of the thesis.

As this paper describes the overall process of the thesis project, the tool artefacts are described in broad terms. In each case, the initial motive or seed for building the tool is described. Then it becomes discussed what kind of thinking and reading it stimulated and what the process of building began to suggest. Particularly the second artefact is a very free-form process of improvisatory exploration.



Figure 1: The first artefact is used to display shape of a view area in a city plan. The work provoked the question of how space becomes understood through representations.

ARTEFACT 1: A VISIBILITY ALGORITHM

The first artefact is a visualization of person's experience of viewing space in motion as a digital sculpture. The artefact is a graphical visualization written in the C programming language, and has its origins in a master level thesis. The shape is derived from accumulating view cone shapes on top of each other. The ensuing form can then be viewed as an on-screen object. Here the goal was to visualize the dynamic nature of experiencing space and stimulate thinking on the role of representations in the design process.

The initial seed of the work was an experience of looking at a statue in a public space. In the middle of a small area, seen through a window, it prompted to think about visibility in that situation. The statue is something that becomes defined through its being seen. What would the statue "see", and what would the shape be like?

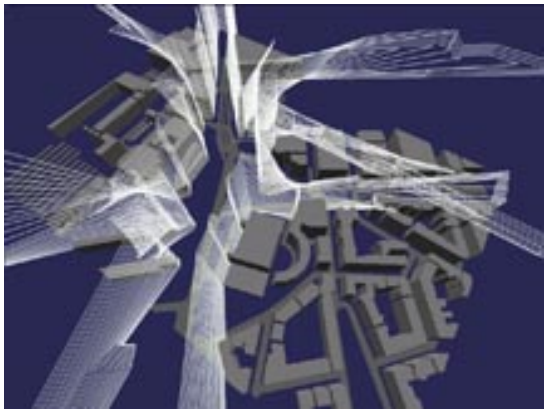


Figure 2: The first artefact became an exhibition piece which illustrates the experience of moving through space as a three-dimensional sculpture made of the view shapes.

This prompted an exercise of drawing the shapes of visibility for that statue in a plan view of the site. Later, this process was made into a computer program which produces these shapes very quickly. The shape could be shown in motion both as an interactive computer animation (Figure 1) and as a time-exposure object (Figure 2).

Later I learned the kind of shapes have been presented by Michael Benedikt (1979) and discussed in space syntax literature. Benedikt also built a similar sculpture by stacking cardboard shapes, and interestingly used point lights to create an interactive way to manipulate the shapes in a model. The research field of space syntax and attempts to quantify the use of the figures was not what felt right for an art and design approach.

The motion sculpture encouraged reading of literature related to perception and experience, such as Gibson's perceptual theory (1986) and more phenomenologically oriented literature, such as work by Norberg-Schulz (1980). It became clear that experience of space is not something that can become exhausted or determined through its geometric description. As a first step in the whole

research project the first artefact helped set the theme for the later tools and some foundational reading on the topic of perceiving space. Forming an opinion on how environments are experienced, the dynamism of movement became one cornerstone for the author's personal theory of space.

Retrospectively, the process can be analyzed in terms of the generative strategy and primary generator, even if these concepts were not consciously known at the time. It is worth to note that the original starting point for the case was someone else's work of art, a statue. The situation with the statue helped collect together thoughts that otherwise might have remained too scattered to become a design project. The initial impetus into thinking visibility was quickly joined by assumptions about using a specific representation (the plan drawing) and a computer program as a device to produce the motion shapes through a mechanical time-exposure. In hindsight, these leaps were somewhat premature, although these choices also provoked questioning of these conventions. For example, the shapes can be easily produced from a city plan, but not from more complex natural and organic environments.

Gibson's perceptual theory and phenomenological approaches would be in agreement in that images of space are powerful. There's no such thing as mere appearance. The use of images and visual tools can be made central to a design process. More realistic images can be used as a virtual world (Schön, 1991, 157) for testing design moves and experimental situations. Drawing generatively, more inspirational images can be brought in into the design process.

Together with the literature, the first artefact established groundwork for thinking about space and set the tone for research. As the focus was on tools, it was practical to keep the general understanding of space as simple as possible. Gibson's idea of grounding meaning of space into a basic perceptual structure provided a sobering backdrop for this.

ARTEFACT 2: A HAND-HELD TOOL

The second artefact is a hand-held device. (Figure 3) This device uses a sensor to measure colours from the environment and feeds the measures into a laptop computer, running another piece of software. The motive was to probe into the idea of tool in as simple way as possible, looking it as a hand held device, like a screwdriver or a wrench. This was partly a counter-move to the first artefact, which had begun to seem an overtly conceptual piece of armchair research. As a change, it was more attractive to go to sites and places and do activities there. After all, the work for the first artefact had also its beginnings in a real site.

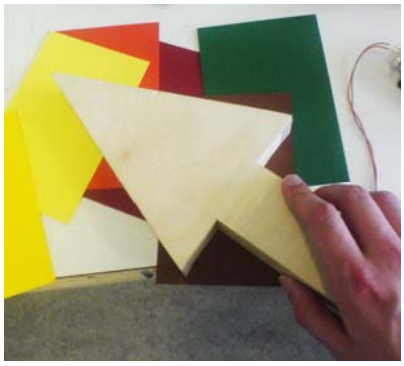


Figure 3: The second artefact is a hand held tool for picking colours from the physical environment.

The practice-led research allows this initial uncertainty or even simple-mindedness in the design stage, as long as the project begins to grow into a productive direction. The process around the second artefact was especially free-form, each successive finding suggesting something else.

A major impetus for the project was a colour sensor, found from a catalogue of electronic parts. The tool was built as a casing around hardware parts that were realistically available at that point. The sensor capabilities were explored by moving the sensor around by hand. (Figure 4) This setup began to suggest different ideas, such as a way of using the device as a physical pointer, using different colours in physical environment for activating functions inside the computer. This gave rise to the shape of the device as an arrow pointer (Figure 3), mostly as a humorous reference to the pointer in a desktop environment. It was found to be a good shape for the casing to contain all the necessary electronics. It also helped settle that the tool would be a hand-held tool.

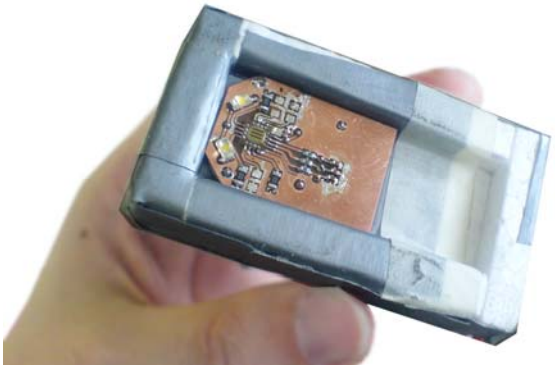


Figure 4: The colour sensor, a 5x5mm piece of electronic equipment (Near the nose of the board), can be considered the original seed of the hand held tool and the approaches that followed from it. The circuit board is designed by Jussi Mikkonen, MSc.

The tool was created in collaboration with an engineer, who built the circuit board for containing the sensor, and also supplied a microcontroller. Here the role of collaborations and available competences also become important in a generative, exploratory mode of design.

The device was taken to outdoor sites, with the intent of collecting sets of colours as an image of a site. This is derived from a common practice of interior designers and architects who may collect colour maps as part of their site survey. Other kind of data collection, such as photography and measurements, were avoided. Only the colours collection would stand as the definite material gathered from the site.

Working outdoors with the device and a laptop proved to be cumbersome. Besides other problems, the light blocking of the casing was inadequate and the tool could not be used at that time. This technical failure of the device prompted me to execute the same task with water colours, without using the digital tool. This unexpected situation made me think how the underlying concept of what I was seeking to do was not dependent on a specific material tool. However, it is important to note that the tool building process preceded this insight.

At the same time as the reliability of the device was improved, it was made into a wireless, autonomous unit. The tool was remade using a readymade Arduino kit and Processing language for more reliability, but the original colour sensor module design remained. The device could be carried to outdoors without needing the laptop. The readings were stored in the miniscule storage of the microcontroller itself.

This version of the tool was used to collect colour sets from various sites, for example in an excursion trip to the Lapland of the north. The building of the tool as a concrete process was also a way to build a relation to the newly encountered sites. Equipped with the lens the tool provided, the journey to the north prompted thinking about the significance of a locality as an origin of design ideas. Architects have traditionally utilized site features as an inspiration and generative basis for their work, as already noted by Darke (1984). Existing geometrical layout, nature and cultural features can serve as an original impetus for a new architectural addition. The collection of colours can serve in this role as well.

The choice to focus on tools as generative guide for design was made during the building and use of this artefact. This part of the process around the tool forms an important step within this practice-led research project. The built tool, its breakdown, subsequent development and the journey to the north ought to seen as aspects of the same story and not easily detachable from each other.

ARTEFACT 3: IMPLICIT STRATEGY IN A TOOL

The third artefact addresses the more implicit generative strategies that become into play when designers or artists work with a medium. When working generatively, drawing and physical materials are arranged with some organizing principle. The paper surface affords certain kind of drawing over others. Drawing, as Klee's example showed, is flexible as it allows the creation of rules. In contrast, computer modelling often offers ready made components, which although can be used generatively, rarely allow very subtle ways of creating ones own rules.

The third artefact is a tool for modelling spaces from little tiles. The tiles are incrementally added to the scene, like a line is drawn on paper. Only here, the point can be moved into the six main axes along the three-dimensional grid. (Figure 6) Walls and volumes

are not offered as components, but have to be "grown" by extending the line in different directions. This deviation from the more usual component-based modelling was meant to allow more diverse self-creation of rules in modelling.

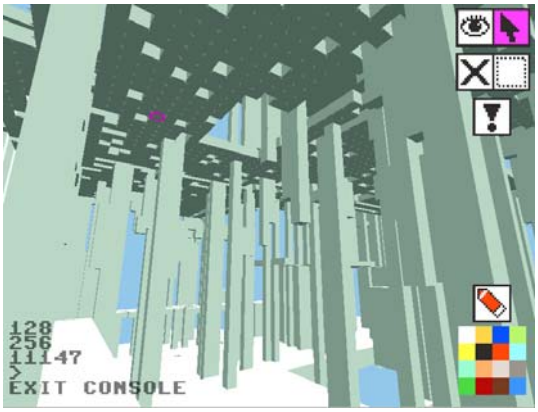


Figure 5: As the third artefact became functional, it could be used to explore semi-random shapes and forms.

This tool was more extensively tested with other people than the first two. As the tool was meant to cap off the thesis project, the result should not suggest yet another artefact.

The first two artefacts helped build a relation to a general idea of space and to a specific site. It felt an appropriate time to address the issue of proposing designs and a more difficult aspect of design tools, namely drawing. The artefact represents an attempt to import elements of drawing to a computer design tool. I felt I was equipped with good enough set of concepts that would allow me to address the more ephemeral and hard to track ways in which drawing relates to design creation.

The first instinct was to consider geometries and the grid as a constraining device, as the idea of constraint had become to the fore when working with the second artefact. As the software enforces a first person view into the modelled space, it also was influenced by the questions of motion and perception that had arisen in the aftermath of the first artefact. Free motion of the viewpoint is integral part of the modelling experience. Another influence was the fluidity of motion in first person video games.

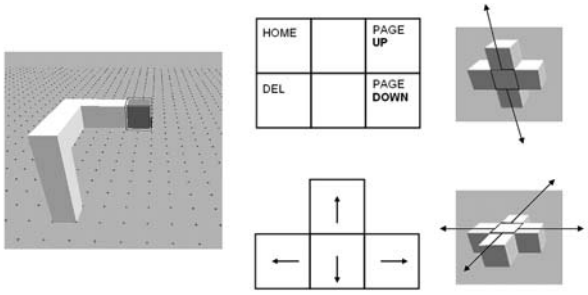


Figure 6: The interaction design of the modeller, an approximation of drawing a line in six directions.

During the making of the software, the software was used to produce preliminary outcomes. (Figure 5) Yet pen-and-paper drawing was used extensively during the designing of the software, as it was necessary to visualize desired outcomes or thorny programming situations. A certain mismatch became apparent. It became clear I could actually draw the kind of spaces I was hoping the program could achieve. (Figure 7) The question then arose what the value of the program was? This could be called a definite moment of surprise, when the process of normal experimentation yields a situation which cannot be immediately explained. It provokes the retrospective "reflection-on-action" as described by Schön (1991)

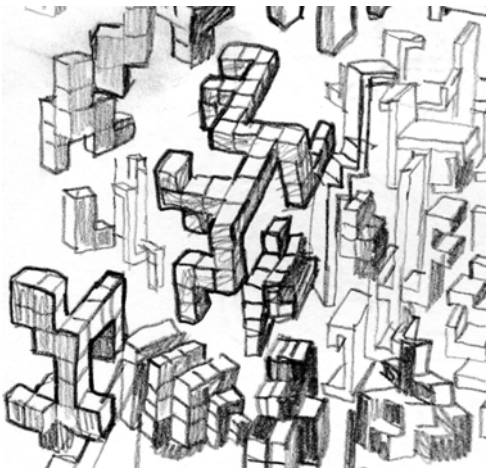


Figure 7: The development of the program heightened aspects of the sketching process.

I concluded that the program had less to do with perceived utility or the possibility of achieving novel form. It was meant to explore space in a way I personally had been unable to achieve through pen and paper drawing. Unknowingly, I had come to set a goal for my sketching that I hoped the creation of the program could help achieve. The making of the tool was a continuation of the examination of the

role of cube and a geometrical rule in a drawing, which was the real chosen generative strategy. This same generative concept is in play in both the drawings and the program.

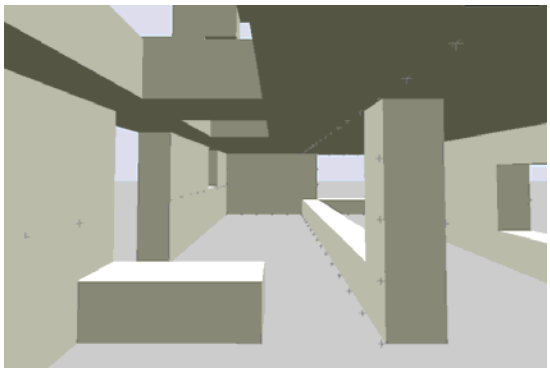


Figure 8: Student work: an interior results as a by-product of making the exterior. After minor adjustments it worked as a sketch of an interior layout.

The drawing process became understood as a set of vaguely imposed rules that constitutes part of the personal theory of space, or a material counterpart of how I believe space ought to be drawn. These rules can be summoned and imposed by will, or even combined at various time during the drawing. In the computer program the idea of making space becomes solidified as a set of rules.

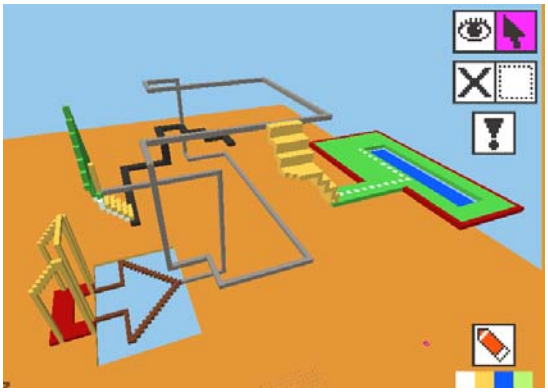


Figure 9: Student work: the way the tool works suggested a design idea based on mazes, motion and games.

Additionally, this tool was tested with other designers. Although these studies can be examined separately, they are here seen as an element of the larger process of practice-led research, as an experiment in sharing ways of making via presenting a tool. 24 designers and design students were each invited to a short session, where they used the software to produce a design doodle or a sketch. The processes were collected in form of logs and video. The material could be used to see how the constrained setting would be coped with, and how and if the tool would suggest design outcomes. Much of the work showed a willingness to merely pass the "test" rapidly, producing a very condensed view into how a generative approach might emerge in a tight situation. A handful of the work shows more definite ways a design idea can become suggested by a tool. (Figure 8 and Figure 9) Some outcomes showed influence from outside the test setting, which was hard to track, but many outcomes were apparently suggested by the software without the tool imposing the approach directly. Looking at the processes of others reminded me of my own chain of exploration in the larger research process and thus provided with clues to the way the entire project could be viewed.

DISCUSSION

I've described three projects of building design tool artefacts. The tools are designed objects, and each has informed the premises of the latter. Focusing on one's own long process allows a very intimate look into the way tool building may become a part of personal theory development and a practice-led research process. Looking back is a matter of tracking influences and the associated thinking related to the different parts.

RETROSPECTIVE LOOK AT THE TRILOGY

The three artefacts each represent an angle to the topic of spatial design tools. At the same time each of the tools set up the premises for follow-ups. The first artefact was an attempt to clarify the experience of moving in space and to build directly a visualization of what I then thought would be a "theory of space". Making the second artefact as a hand held tool became a way of breaking this notion. The design tool was explored instead as a physical object for building a connection to a place. The third artefact began as an attempt to make a novel kind of modelling software, but became understood as a means of focusing into the ongoing process of design drawing.

The overall project demonstrates the practice-led research approach. The artefacts are initially openings to a topic, but are also valid parts of the

progression of reflective thinking. Each tool was different, but on a more general level the act of theory-building becomes repeated, merely viewed in new light. The overall process of tool building is a way to bring cohesion, deliberation and organization into one's own personal theory building. The lessons learned from the artefacts could also be fed back into the more established habits of drawing and modelling. This is how to engage in Schön's reflective research, a purposeful activity that enhances the process of reflection-in-action, exercised in the medium of sketching.

REFLECTIONS ON THE METHOD

The practical design has indeed "led" the research, but not haphazardly. The overall research topic had from first been established as visual, conceptual design tools. The question was how design tools can be built by designers and what it might benefit them. These themes kept the process together, even when later focusing more on the building of personal theory and generative strategy. A well defined research question or problem area is helpful, but in reality it is also redefined throughout the project.

Putting other designers to use the tools, as a complementary approach, has also provided insight. These situations put into question how the tools might transfer knowledge or suggest ideas to others. As the tools are personal, they can elicit interest and fascination in some, but easily arouse suspicion and reluctance from others. This suggests people have very diverse approaches to such tools. The tools alone provoke thinking and critique, just as design objects or artworks would. The combination of text and the design outcomes can pass on more definite knowledge others can relate to.

In this practice-led project, the thesis project became an insider account of the ways a designing researcher builds tools. This process has much subjective elements in it. Any other designer might have found different routes from the initial starting points. Challenges arise when separating the more subjective elements from that which is generally useful, inspirational or replicable by others. Repeating a process of building a tool in three different ways has offered enough material to provide some overview into what to share about such a process.

A recurring occurrence during the course of project was that as an idea came into being, it was first built in a very technical way. Later it was possible to make use of the idea without these training wheels. Each concrete design artefact became a stepping stone also in this sense. An idea about the perception of space was built into a digital sculpture, but later remained a conceptual idea influenced by readings into literature. The hand-held digital device broke down and showed that ultimately the same process could be achieved in traditional means, provoking the thinking on the underlying concept of a designer's tool. The third artefact derived ideas from drawing into modelling software, but turned out to be more instrumental in self-learning drawing.

This became a central element of the method. Building a new tool highlights the way designer engages with a material or a conceptual idea. At first the idea is made simple, through its technical implementation, but later grows to be a more conceptual understanding of the topic, which can be summoned in its more rigid form if necessary. The tool building is an act of embedding personal theory idea in the tool. It is also a way to establish cohesion to the research project. The artefacts anchor topics the researcher becomes committed to.

TOOL BUILDING AS KNOWLEDGE PRODUCTION

All the tools here are thematically connected to the idea of personal theory about space. Some kind of personal theory, conscious or not, whether called a design philosophy or an artistic credo, is necessary for making proposals about form and space. It is not a theory in a hard scientific sense, but more akin to the kind of belief that a designer or artist upholds and subscribes to.

Design drawing skill is an example of practical knowledge, which can in part be explained in a visual and textual form. Drawing can be used generatively, as the examples provided by Paul Klee (1961) already demonstrate. A point on a paper becomes a line when the pen is moved. Further organization of the

lines into planes and volumes presupposes a belief into the utility or expressiveness of such shapes. These beliefs are not in the tool, but developed and learned by the artist.

Perspective method manuals for designers stress that learning the rigid method is merely a way to learn free hand drawing, which is more useful in actual designing. (Doblin, 1950)(Lockard, 1982) This skill is achieved through learning to draw principal elements, such as cubes, in freehand. In this way the books supply one model for the reasoning that may go with the pen, still allowing the drawing to lead.

If the perspective method books are able to deliver meaningful points about design drawing, the books might also be seen as a rudimentary model for distributing personal theory and knowledge as tools. After all, the manuals were created by practicing designers and modified along to suit particular needs of the different design disciplines. Also, the authors refer to and build on past work.

To be of wider application, this idea requires a framework for discussing conceptual tools and not merely drawing devices. Practice-led research into tools can serve as a model for engaging into similar activity across wider range of topics. In light of the theoretical literature and the experiences I've had, the ability to deploy and manage generative strategies seems to describe an important aspect of what designers, artists and craftsmen do. Especially this is so when they engage into creative processes with materials, tools and mediums.

The skill of design in one part is the use of these strategies. The concept of generative strategy helps get further into explaining moves that might become hidden under the terms of inspiration, intuition and influence. The adoption of a generative strategy is not always conscious, but often the forking of choices and abandoned routes can be identified afterwards. The focus on generation and the generative strategy has been useful in reviewing one's own past design activity from a consistent angle. It is one possible concept for analysing and reporting the genesis of tool artefacts.

NOTES ABOUT COMPUTER USE

Interactive software can be used to show and demonstrate things. The demonstration can influence or set a precedent for how to work in a different medium altogether. For example, an algorithm might show geometries and shapes that the designer could not have found through drawing. This does not mean that the computer is then essential from that point on. In fact, the same shapes can then be replicated in drawings. In Sevaldson's work, the screen was considered one potential source of design inspiration, regardless of what the program was intended for.

Here it has been offered that the rigidity computers sometimes introduce to design may be initially helpful, even if the tool became later abandoned. The first artefact was a mechanistic view into the complexity of the experience of motion and the latter deliberately treated the locality of a site as a set of measures. This definiteness helped gain an entry point for a learning experience.

Rapidness is an important aspect of tools, and relates to the issues of sensitivity and gestures Hummels raised in her work. This rapidness is not the speed of efficiency, but the speed that enables different generative approaches. For example, rapid brushstrokes are an enabler of stylistic outcome for an impressionist or expressionist painter, not a way to produce a high quantity of paintings. If software does not make this rapidness possible, its expressiveness becomes severed in this direction. The third artefact and its use is an essay into the examination of a three-dimensional "brushstroke" of the cubic volume, played against paper-and-pen drawing.

Conversely, some tools are inherently slow, and do not represent an immediate bodily skill. A computer artist might acquire a dataset, visualize it in a program, transform the data set and produce a new type of visualization. This is the larger picture of the kind of processes Sevaldson describes, examined through the second artefact. This takes time, and the way of working is different from brushes or clay.

In this thesis work both the rapid, tactile and the slow construction of tools have been utilized. I did not want to explore design tools either solely based on intuitive tactile relation or as distant, interactive representations. Both are used in a creative process.

CONCLUSIONS

This paper has discussed making design tool artefacts, originating from a designing researchers practical work. It has reported on a long chain of design tool artefacts as a process, where each tool and its use informed reading of theory and the subsequent tools. This process is seen as part of the designer's personal theory building, a base from which generative ideation stems from. The process is a demonstration of one way how a designer might cultivate his or her own personal theory through the building of tools.

One model offered here is that the engagement into a tool building is a kind of self-imposed contract. Because the tools are elevated into designed objects themselves, the artist or designer is more inclined to continue and finish the tool, at the same time building an understanding of its significance. During this process learning and insight occurs. When discussing more flexible, ephemeral tools like drawing, it is not easy to delineate the extents of such a contract. Building the tools into definite objects distinctly outlines different aspects of the ongoing process and helps reflective thinking.

The practice-led researcher has the responsibility to transfer otherwise tacit and implicit modes of knowing to shareable knowledge. This is achieved mainly about writing about the done work. Part of the knowledge that is exhibited in the making is more akin to craft skills and cannot be immediately shared. The more overarching process becomes the replicable element that can be emulated or compared. Others need not attempt to build or use the particular tools presented here, but to consider the possibilities of what such a building process might entail.

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